ENGR 102 Summer Bridge
Day 2 Instructor Key

Variables, Arithmetic, Conversion, and Formatted Output

20 points • approximately 20-25 minutes

Name		UIN		Section	

Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace quotient, remainder, and final state. 4 points

```
minutes = 367
hours = minutes // 60
remaining = minutes % 60
code = hours * 10 + remaining
print(hours, remaining, code)
```

Show each arithmetic result and give the exact output.

Answer and explanation

367 // 60 is 6 and 367 % 60 is 7. The final expression stores 6 * 10 + 7 = 67. Exact output: 6 7 67.

Question 2. Distinguish text combination from numeric addition. 4 points

```
raw = "8"
extra = 3
print(raw + str(extra))
print(int(raw) + extra)
```

Name the type used by each + operation and give both exact output lines.

Answer and explanation

The first + combines two strings, so it produces 83. The second converts raw to integer 8 and performs numeric addition, producing 11. Exact output lines: 83 and 11.

Question 3. Repair a missing input conversion. 4 points

```
kits = "4"
total = kits * 6
print(f"Total items: {total}")
```

Give the actual output, identify the stored type that caused it, and write the minimal repair.

Answer and explanation

kits is a string, so multiplying by 6 repeats the character six times. Actual output is Total items: 444444. Repair with kits = int(kits) before the calculation, which produces Total items: 24.

Question 4. Program construction — 8 points

Write one complete Python program that satisfies every requirement.

- Read length and width as floating-point inputs.
- Compute rectangle area and perimeter.
- Print Area: followed by one decimal place, then Perimeter: followed by one decimal place.
- The program must work for values other than the example.

Answer and explanation

```
length = float(input())
width = float(input())

area = length * width
perimeter = 2 * (length + width)

print(f"Area: {area:.1f}")
print(f"Perimeter: {perimeter:.1f}")
```

Both input results are text until float performs conversion. The calculations use the converted variables.

For inputs 3 and 5, area is 15.0 and perimeter is 16.0, so the required output lines are Area: 15.0 and Perimeter: 16.0.

Scoring guidance

2 input/conversion, 2 named calculations, 2 correct formulas, 2 exact f-string labels and precision.